

Upper White flood-map completion: interview note

HUC-8 05120201. Methods and claim discipline. P(sfha | hydro) on a 30 m EPSG:5070 grid.

Revisions: 2026-08-27: first interview note. 2026-08-27: Figures 1 to 3 (basin disagreement, five office-to-max zooms, two 2008 reaches) and Figure 4 (five-site Indiana 2025 parcels). D tables unchanged. Generated 2026-08-28T02:38:26Z. Git fca780871ef2. Product artifacts at c0e9d5e. Parcels tag: parcels-five-sites.

Abstract

P(sfha | hydro) is a map-completion score, not a 1%-annual-chance exceedance probability.

Calibrated OOF PR-AUC (0.36) beats the SFHA-rate baseline (0.10) and a HAND score (0.24).

Five Zone X plants have one wet cell in a 120 m window; site-mean P is 0.06 to 0.19; none clear 0.50 on the footprint.

THURSDAY POOLS is the only large-inventory terrain hit, and it is neighboring land.

June 2008 Appendix 2 does not cover the industrial core of 05120201.

To three decimals: raw PR-AUC 0.369, calibrated 0.362, Brier 0.073 after isotonic calibration. Stage D samples p_sfha_calibrated.tif only. Do not treat 0.75 as a percent chance on the raw raster.

Table 1. Five Zone X plants ranked on window-max P

Plant	2023 on-site release (lb)	Highest P in 120 m	Mean P in 120 m	What the high cell is
THURSDAY POOLS	257590	0.769	p_mean 0.152	neighboring land, unshaded X
FGF LLC	27335	0.780	p_mean 0.060	adjacent hydro (floodway)
ROYAL SPA CORP	4950	0.774	p_mean 0.113	adjacent hydro (waterbody)
LINDE GAS & EQUIPMENT	1048	0.763	p_mean 0.192	neighboring land, unshaded X
MAGNA POWERTRAIN EAST	0	0.789	p_mean 0.098	neighboring land, unshaded X

Table 1 ranks on the wettest 30 m cell inside a 120 m window. Mean P is the footprint. At THURSDAY POOLS (p_mean 0.152) that cell is neighboring unshaded X, HAND = 0, 120 m from the office point: an edge screen, not a wet footprint. FGF LLC (p_mean 0.060) and ROYAL SPA CORP (p_mean 0.113) are adjacent-hydro footnotes (floodway corner; waterbody). MAGNA POWERTRAIN EAST (p_mean 0.098) is in the list because the rank is window-max P. Inventory pounds are the table column only.

Figure 1. Basin disagreement

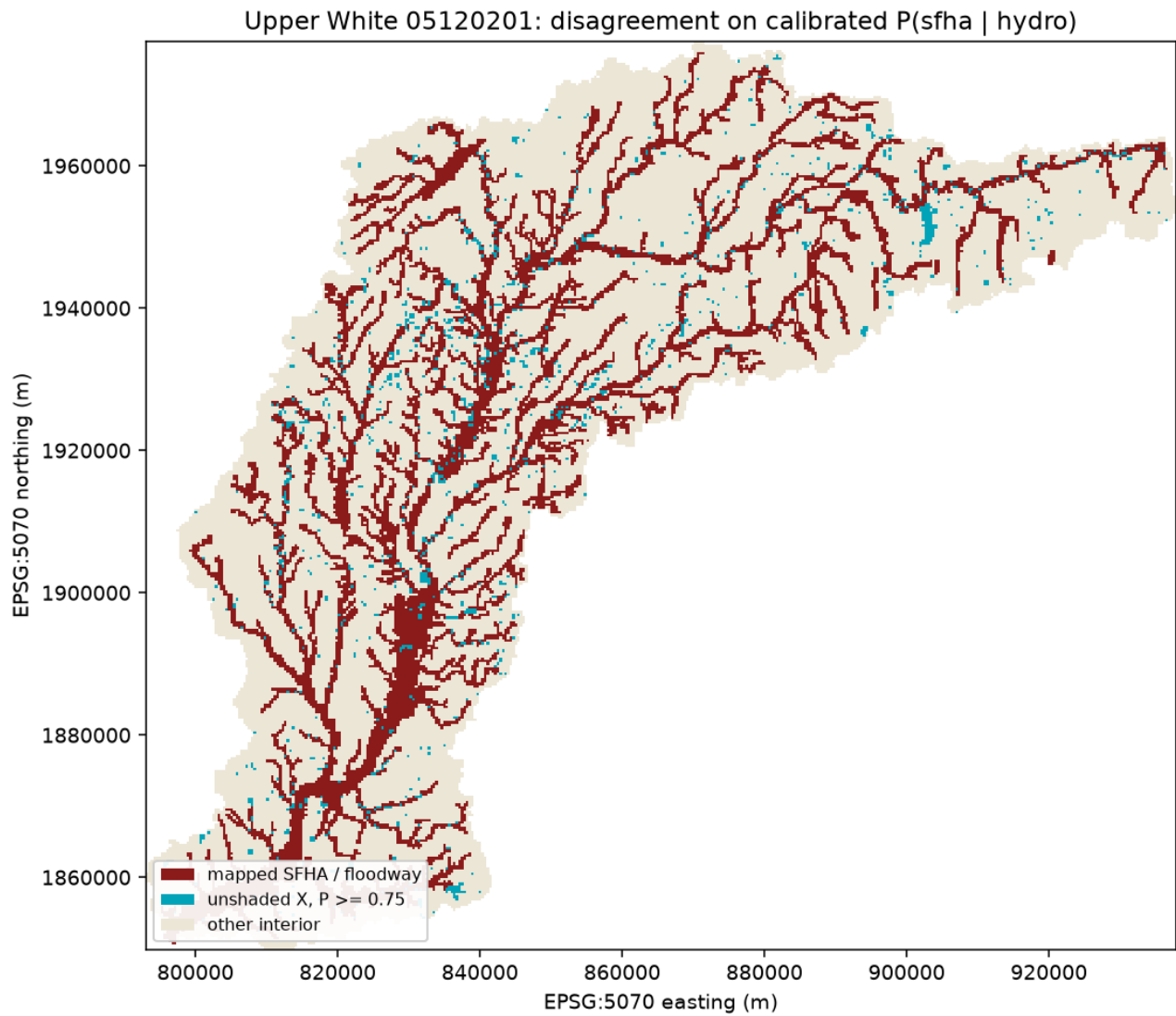


Figure 1. Basin disagreement on calibrated $P(\text{sfha} \mid \text{hydro})$. Dark red: mapped SFHA and floodway. Cyan: unshaded X with calibrated $P \geq 0.75$ (same t as Table 1 window-max, pixel not plant). Pale: other interior. Cyan is map-completion on the FIRM, not a plant-level hazard list.

Figure 2. Five office-to-max windows

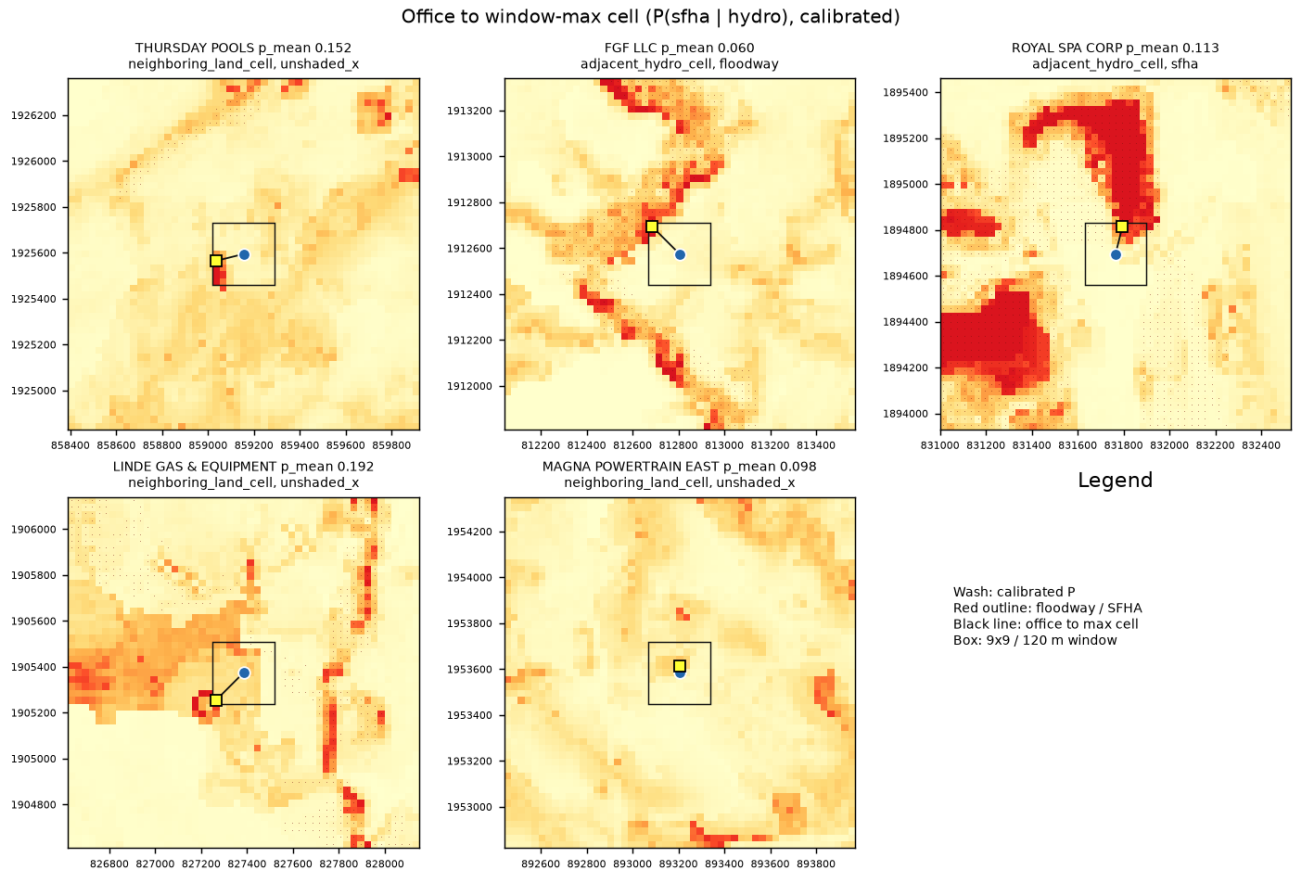


Figure 2. Office point to window-max cell. Each panel title has p_{mean} . Wash is calibrated P . Box is the 9x9 (120 m) window. THURSDAY POOLS $p_{\text{mean}} 0.152$ is neighboring unshaded X. FGF LLC $p_{\text{mean}} 0.060$ and ROYAL SPA CORP $p_{\text{mean}} 0.113$ are adjacent hydro. MAGNA POWERTRAIN EAST $p_{\text{mean}} 0.098$ ranks on window-max.

Figure 3. Two 2008 reaches

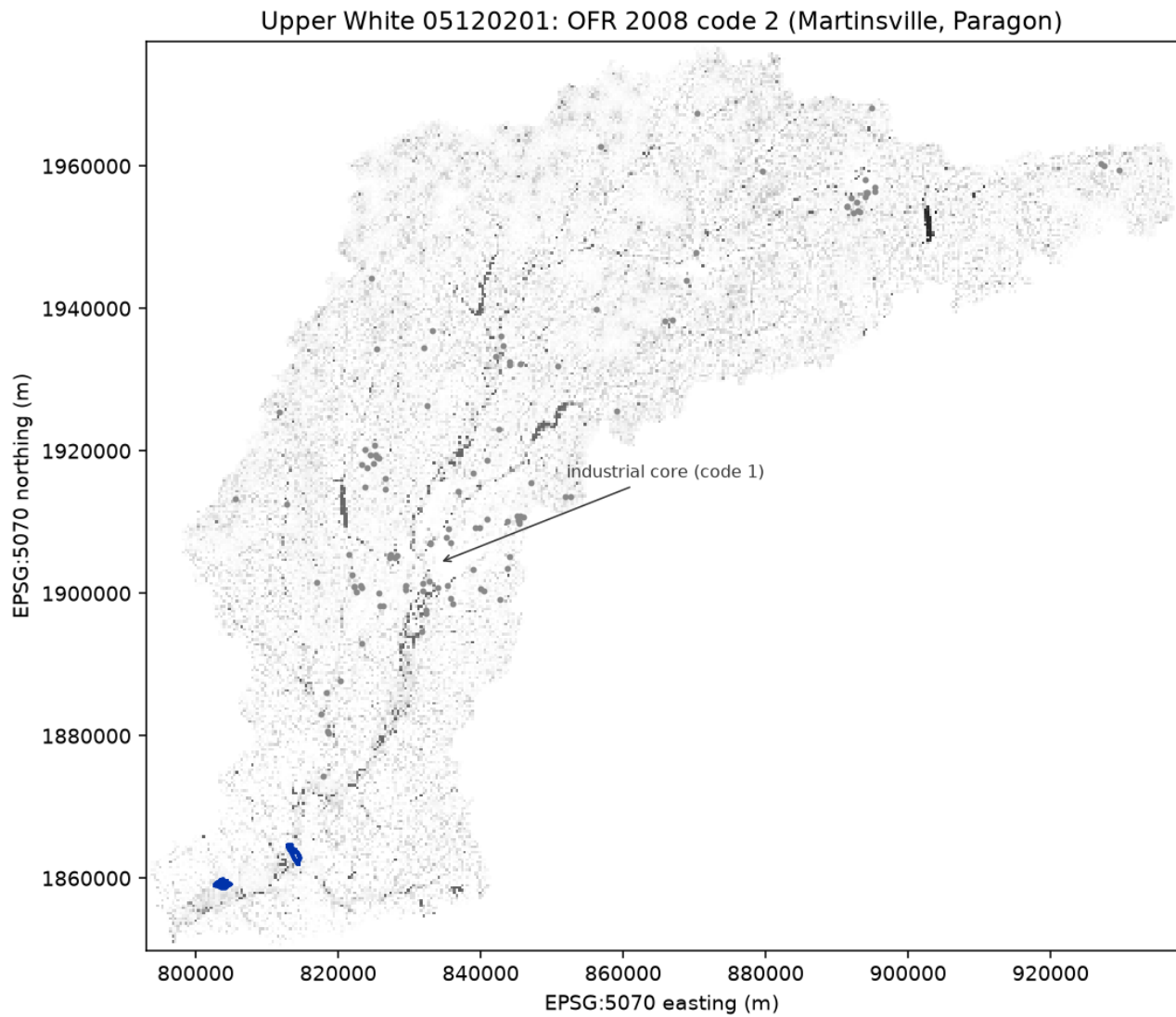


Figure 3. June 7-9, 2008 inundation (OFR 2008-1322) code 2 on the same HUC. Blue polygons: Martinsville and Paragon. Grey dots: 117 TRI office points. D2 stays 117 / 0. Appendix 2 is reach-scale; the industrial core is code 1.

Figure 4. Five-site parcels

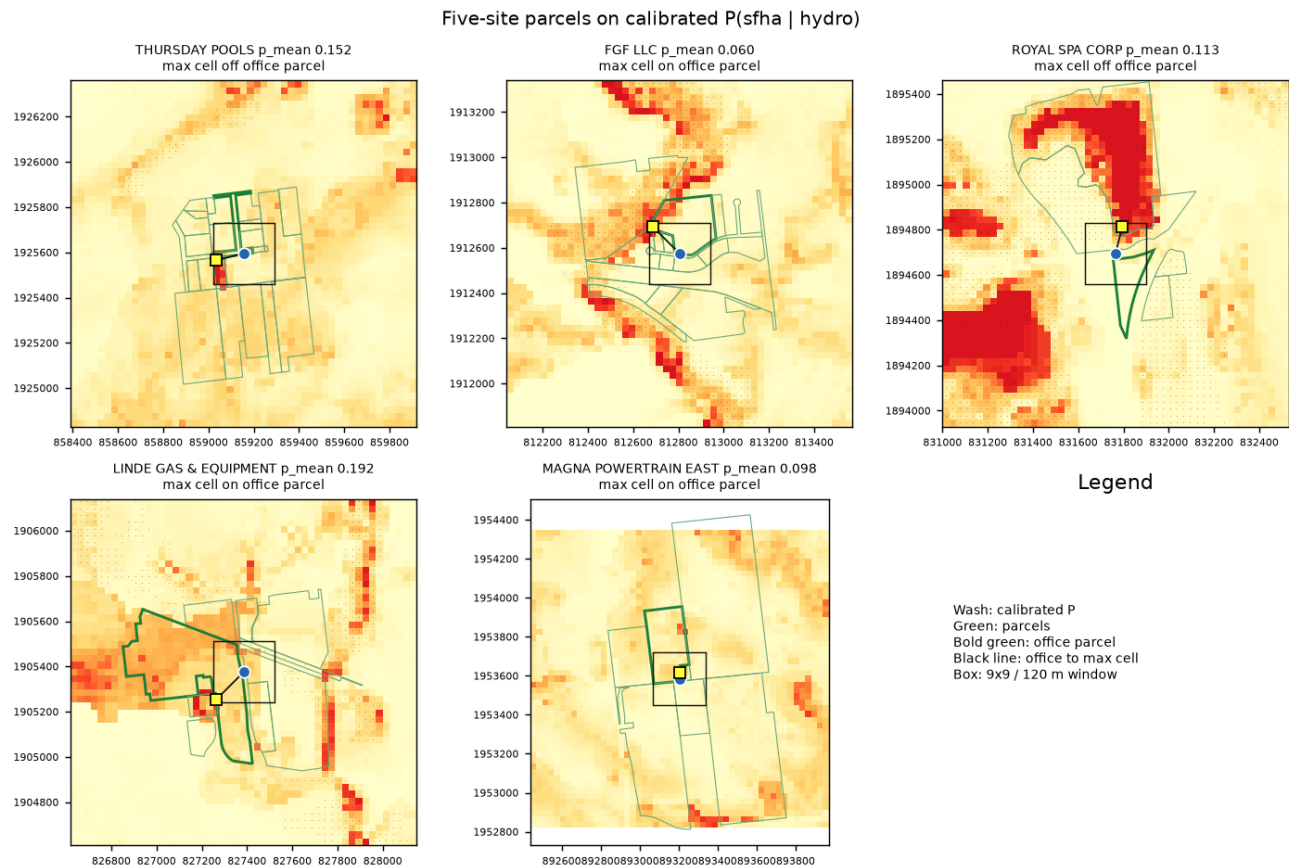


Figure 4. Indiana 2025 parcels (GIS Data Harvest) on the five Table 1 sites only. Snap 30 m. D tables unchanged. THURSDAY POOLS $p_{\text{mean}} 0.152$: max cell off the office parcel. FGF LLC $p_{\text{mean}} 0.060$: floodway cell on the office parcel. ROYAL SPA CORP $p_{\text{mean}} 0.113$: waterbody cell off the office parcel. LINDE GAS & EQUIPMENT $p_{\text{mean}} 0.192$ and MAGNA POWERTRAIN EAST $p_{\text{mean}} 0.098$: max cell on the office parcel. Folder: logs/parcels/.

What you can do with it

1. Portfolio (methods and claim discipline).

This is the highest-value use. Geospatial, climate-risk, insurance, and civic-tech readers can see that the work locked a geography (Upper White 05120201), refused a bad label (IndianaMap dropped unshaded Zone X; NFHL layer 28 with where=1=1 restored it), calibrated probabilities instead of waving at 0.75, split window-max from site-mean so five factories were not called flooded, and wrote claim bans into CI. Lead with the five abstract sentences, Table 1, and Figures 1 to 4. Do not pitch this as hidden flood zones.

2. Interview talk (about 90 seconds).

I trained a model to complete FEMA floodplain maps in the Upper White basin. It beats a HAND baseline on ranking. When I overlay TRI plants, five Zone X sites have one wet cell in a 120 m window. Site-average probability stays low. The 2008 USGS inundation maps do not cover Indianapolis industry. So this is map completion plus an edge screen, not a plant-level hazard list.

If they push: calibration, spatial CV, and why Zone X extraction was the actual bug. Those notes are below. They are methods, not a second finding.

3. This short note.

Five sentences as abstract, Table 1, Figures 1 to 4, one limitations paragraph. That is the sendable object. The repo is martialsystems/indiana_flood_completion (private). Parcels tag: parcels-five-sites.

Stage C metrics

Metric	Value
PR-AUC, quoted	0.36 (raw 0.369, calibrated 0.362)
SFHA-rate baseline	0.10 (0.097)
HAND score PR-AUC	0.24 (0.243)
Brier after isotonic	0.073
Brier, raw scores	0.177 (constant classifier 0.088)
CV	leave-one-HUC-10-out, 17 watersheds, 1-pixel halo
Features	slope, TWI, HAND, dist_flowline, dist_waterbody, NLCD impervious
HSG in model	no
D sample raster	p_sfha_calibrated.tif
In-HUC TRI 2023	117 (101 unshaded X; 16 already mapped hazard)

Raw booster scores ranked and were miscalibrated (Brier worse than the constant). Isotonic regression on out-of-fold scores, same HUC-10 cuts, wrote the calibrated raster. Rank held (0.369 to 0.362). D never samples raw p_sfha.tif.

If they push

Zone X extract: IndianaMap FIRM 2023 omitted AREA OF MINIMAL FLOOD HAZARD polygons. Monument Circle, a Carmel till-plain cell, and a Delaware field painted as unmapped. D1 is zone_class == unshaded_x, so that extract would have dropped Indianapolis mapped Zone X from the overlay. FEMA NFHL MapServer layer 28, where=1=1, is the live FIRM. Interior unshaded X is 6,986,426 cells (89.23%).

Spatial CV: 17 HUC-10 blocks, leave-one-out, no test HUC-10 in train, including a 1-pixel halo. Labels are binary SFHA (floodway included). HAND-nodata cells are excluded and stay nodata on both P rasters.

Max versus mean: buffer-max is one 30 m cell in a 120 m window. Buffer-mean is the footprint. Zero headline rows have mean P at or above 0.50. Adjacent hydro, after the five-site parcel clip: FGF LLC (p_mean 0.060) floodway cell is on the office parcel; ROYAL SPA CORP (p_mean 0.113) waterbody cell is off the office parcel. THURSDAY POOLS (p_mean 0.152) max cell is off the office parcel.

Claim scanner: floodmap.claims fails a report that uses banned tokens, treats P as a 1%-annual-chance exceedance, prints empty D2 without the code-1 / code-2 split, or names a D1 headline plant without p_mean in the same window. GraphForge pin: whiteforge/.

Close-out: SHAP

Global SHAP on C features, no HSG: HAND first (mean |SHAP| 1.37), then distance to water, then slope. TWI is last because HAND already ate the wetness signal. At the THURSDAY POOLS max cell (p_mean 0.152), HAND = 0 is the local driver and the cell is still unshaded X. Write-up, not discovery.

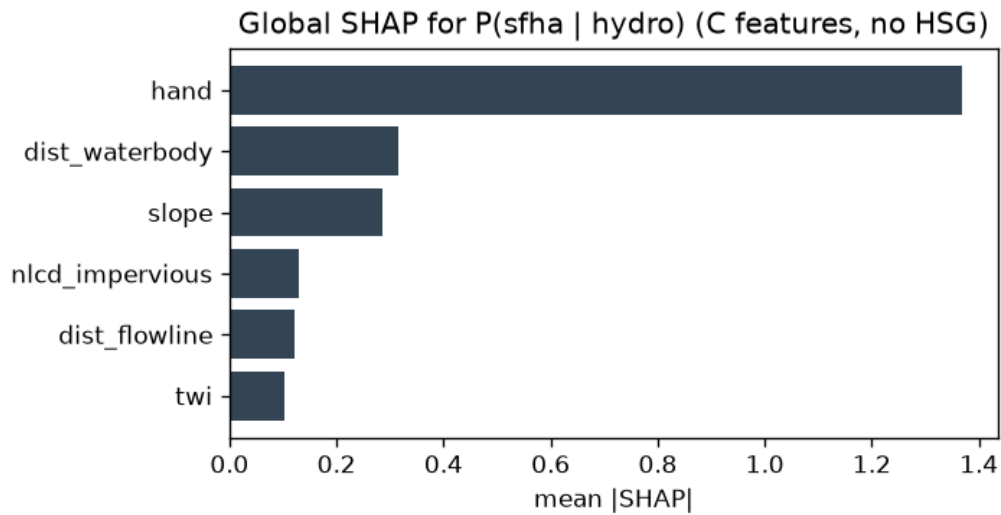


Figure 5. Global SHAP for C features. Close-out, not a new result.

Limitations

Parcels: five Table 1 sites only (Indiana 2025 harvest, logs/parcels/). Rest of the HUC has no cadastral clip. No soil: hydrologic soil group is not in this model; gSSURGO C2 was not required for the claim graph. Not a FIRM: P does not replace the effective flood map. 2008 is coverage: 117 plants in mask code 1, 0 in code 2; Appendix 2 reaches in the HUC are Martinsville and Paragon only. The map HTML was not opened in a browser for this note; Figures 1 to 4 are drawn from the calibrated raster, D tables, and the five-site parcel query.